Hossein Aqasizade

Website | Google Scholar | LinkedIn | GitHub

Graduate Research Assistant, Department of Computer Engineering,
University of Mazandaran, Babolsar, Iran

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Educational Background

Master Degree (2020 - 2023)

M.Sc., Computer Engineering, Computer Networks
Department of Computer Engineering, University of Mazandaran, Babolsar, Iran
Thesis: Combining Virtual Machines and Containers in a Cluster
Supervised by: Dr. Ehsan Ataie
Total GPA: 3.88/4 (17.92/20 in Iranian Scale)

Bachelor Degree (2016 - 2020)

B.Sc., Computer Engineering, Information Technology
Department of Computer Engineering, Azad University, Mazandaran, Iran
Final Project: Implemented management system with virtualization
Advised by: Dr. Habib Esmaeelzadeh Rostam
Last two years GPA: 3.17/4 (15.9/20 in Iranian Scale)

Experiences

Teaching Assistant

- **Cloud Computing** (for Master Students), Supervised by Dr. Ehsan Ataie — *Fall 2022*
  Training (Kubernetes, Docker, Prometheus, and Grafana), and Project Management
- **Cloud Computing** (for Bachelor Students), Supervised by Dr. Ehsan Ataie — *Fall 2022*
  Training (Kubernetes, Docker, Prometheus, and Grafana), and Project Management
- **Performance Evaluation** (for Master Students), Supervised by Dr. Ehsan Ataie — *Fall 2021*
  Training, Exercise Definition, and Project Management

Research Assistant

- **Serverless and Machine Learning,** Supervised by Dr. Meysam Roostaie — *Sep 2023 - Present*
  As team leader, I focus on evaluating serverless platforms like Open-FaaS, Fission, Nuclio, Knative, and OpenWhisk in different kubernetes deployments.
- **Optimizing Kubernetes Performance on Xen VMs,** Supervised by Dr. Ehsan Ataie — *Sep 2023 - Present*
  Studying the performance of serverless functions with OpenFaaS and MySQL on Kubernetes in virtualized environments, including Xen VMs in a CloudLab
- **Performance Analysis of Serverless Platforms, Supervised by Dr. Ehsan Ataie** — *Nov 2022 - Aug 2023*
  As team leader, I focus on evaluating serverless platforms like Open-FaaS, Fission, Nuclio, Knative, and OpenWhisk. Our project aims to assess their performance in diverse Kubernetes setups and conditions.

IT Support Specialist, Almas, Mazandaran, Iran — *2017 - 2020*

Research Interests

- Serverless Computing
- Machine Learning
- Deep Learning
- High Performance Computing
- Parallel Computing
- Virtualization

Journal Papers

- Performance Analysis of Containers Running on Top of Virtual Machines *Submitted–2023*

  Description: The performance of running containers within virtualized environments.
  Journal: Cluster Computing
  Authors: ***Hossein Aqasizade***, *Ehsan Ataie, Mostafa Bastam*

- A Comparative Study of Virtualization: Heavyweight Virtualization or Lightweight Virtualization? *Submitted–2023*

  Description: Conducting a comparative analysis to evaluate the performance and resource utilization of heavyweight virtualization and lightweight virtualization techniques.
  Journal: The CSI Journal on Computing Science and Information Technology
  Authors: ***Hossein Aqasizade***, *Ehsan Ataie, Mostafa Bastam*

- Optimizing Kubernetes Performance in: A Study with OpenFaaS on Xen VMs *In preparation*

  Description: Performance of Kubernetes with runtime: OpenFaaS and MySQL over Xen VMs in CloudLab.
  Authors: ***Hossein Aqasizade***, *Ehsan Ataie, Mostafa Bastam*

- Performance Evaluation of Kubernetes Deployments Approaches with OpenFaaS Integration *In preparation*

  Description: Investigating the performance implications of integrating OpenFaaS with different deployment approaches in Kubernetes clusters.
  Authors: ***Hossein Aqasizade***, *Ali Shohadaei, Mohammadreza Pooshani, Ehsan Ataie*

Skills

- Deep Learning (certificate)
- Machine Learning (certificate)
- MLOps (certificate)
- Git & Version Control (certificate)
- Kubernetes (certificate)
- Docker (certificate)
- LPIC 1 and LPIC 2
- LPIC 304 (certificate)
- Python Data Structure (certificate)
- Python Data Science (certificate)
- AWS ECS (certificate)
- CloudLab
- NetConf, NetFlow, NetSNMP, NetXMS
- Prometheus, Grafana, and Zabbix (certificate)
- Innovative Teaching with ChatGPT (certificate)
- Prompt Engineering for ChatGPT (certificate)
- Microsoft Certificate 70-698
- Microsoft Certificate 70-410 (certificate)
- Microsoft Certificate 70-411(certificate)
- Benchmarking Tools (PTS, K6, and ...)
- HPC (High Performance Computing)
- Virtualization (Xen, ESXi, KVM, and ...)
- Serverless Platforms (OpenFaaS, Nuclio, and ...)
- HTML, CSS, JS
- CompTIA Network+ (certificate)
- Leadership and Teamwork (certificate)
- Adaptability and Problem-Solving

Educational Projects

- Implemented **Genetic Algorithm** for feature selection in Mobile Price classification using **Python**
- Implemented a **Neural Network** for Dry Bean classification, utilizing a machine learning framework using **Python**
- Configured **NetSNMP** Configuration for Network Monitoring
- Configured **NetConf** Configuration for Network Device Management
- Configured **NetFlow** Configuration for Traffic Analysis
- Configured **NetXMS** Configuration for Network Monitoring
- Implemented and simulated cloud computing environments using **CloudSim** toolkit in **Eclipse** IDE
- Designed and deployed a queuing system simulation **(M/M/1/K)** using **Python**
- Developed a Python program for a **graph-based access control system** using **Python**